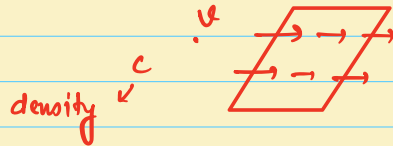


PARTIAL NON-LINEAR INTEGRO DIFFERENTIAL EQUATION

Electrochemical Cell:-

$$\text{Flux} = \text{velocity} \times \text{density} \quad \text{—————} \quad (1)$$



How many particles will pass through the frame in 1s?

In 1s, particles that are v m away, will pass through the plane.

Total volume that passes $\Rightarrow v \times 1 \text{ m}^2 = v \text{ m}^3$

Total amount $= cv$

$$\text{Force} = (-)\text{ve gradient of potential} \quad \text{—————} \quad (2)$$

$$\rightarrow D_i = \frac{RT}{6\pi\mu R_s} \quad \text{—————} \quad (3)$$

$$\mu_i = \mu_i^\circ + RT \ln c_i$$

Electric Potential $= \phi$

$$\text{Force} = -z_i \frac{d\phi}{dx} F$$

charge

Faraday's Constant

(to convert it into mole terms).

Now the force = drag force

$$\rightarrow -z_i F \frac{d\phi}{dx} = 6\pi R_s \mu \cdot v \rightarrow \text{migration velocity}$$

Flux = Diffusive Flux + Migration Flux

$$\rightarrow u = \frac{-z_i F \frac{d\phi}{dx}}{6\pi R_s \mu}$$

substituting from eq'n (3)

$$u = \frac{-z_i F (d\phi/dx) \cdot D_i}{RT}$$

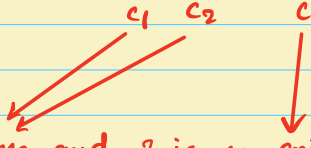
$$\text{Migration Flux: } u \cdot C_i = \frac{-z_i F d\phi/dx \cdot C_i \cdot D_i}{RT}$$

$$\text{Total Flux: } \frac{-z_i F C_i D_i}{RT} \frac{d\phi}{dx} - D_i \frac{dC_i}{dx}$$

Note: In electrical cell, we can use C_i for density as well.

Diffusion - Migration Transport in a System

we have:

- 1) Steady 1D system
 - 2) Potentiostatic Electrolysis
 - 3) Three ionic species
 - 4) Species 1 & 2 are cations and 3 is an anion
 - 5) Only species 1 reacts at the electrode.
- $C_1 \quad C_2 \quad C_3$ - species


Hence for non-reactive species, $N_i = 0$ at SS

At $N_i = 0$

$$-D_i \frac{dC_i}{dx} - \frac{z_i C_i D_i F}{RT} \frac{d\phi}{dx} = 0 \quad i = 2, 3$$

Now,

Galvanostatic Condition ↗ constant current
 i : current density (Amp/m^2).

$$\left(-D_1 \frac{dC_1}{dx} - \frac{z_1 C_1 D_1 F}{RT} \frac{d\phi}{dx} \right) F 2 = i$$

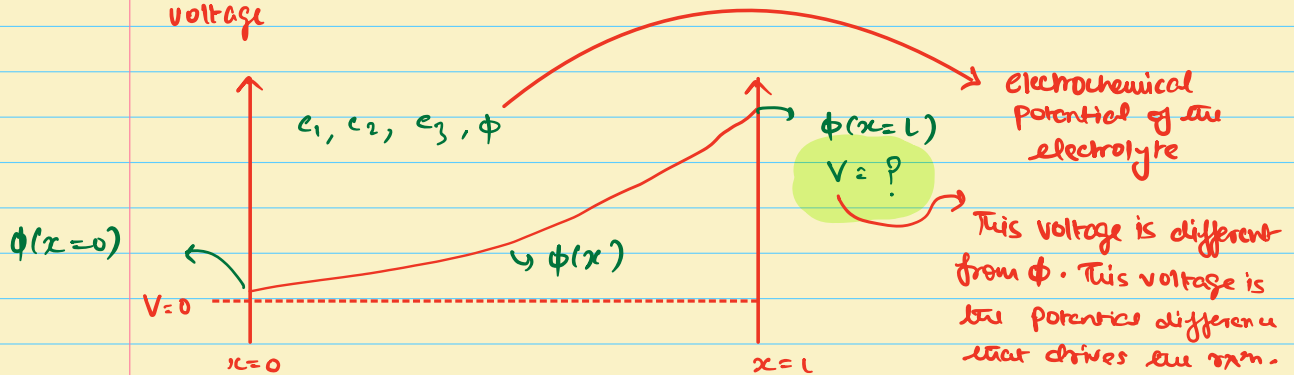

 Molar Flux

now, local neutrality

↳ Approximate law of Physics.

$$\sum z_i c_i = 0$$

→ Now, to keep the cell at constant current, you don't know the voltage



The kinetics of this reaction is governed by the Butler-Volmer kinetics

$$i = i_0 \left\{ \exp \left(-\alpha z_1 \frac{F}{RT} \phi(0) \right) - \frac{c_1(0)}{c_1^0} \exp \left((1-\alpha) z_1 \frac{F}{RT} \phi(0) \right) \right\}$$

0.5 in our case

$c_1(x=0)$

ϕ at $x=0$

reference conc'n

CATHODE

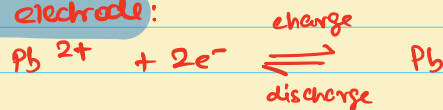
$$i = i_0 \left\{ \frac{c_1(L)}{c_1^0} \exp \left((1-\alpha) z_1 \frac{F}{RT} (\phi(L) - V) \right) - \exp \left(\alpha z_1 \frac{F}{RT} (V - \phi(L)) \right) \right\}$$

ANODE

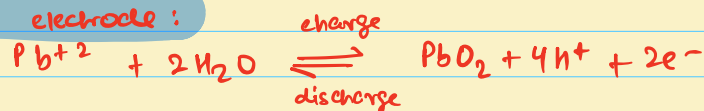
(These are differences of forward and backward reactions)

Reactions:

-ve electrode:



+ve electrode:



Unsteady Mass balance eq'n for i^{th} species:

$$\frac{\partial c_i}{\partial t} = -\frac{\partial N_i}{\partial x}$$

$$\frac{\partial c_i}{\partial t} = D_i \frac{\partial^2 c_i}{\partial x^2} + \frac{z_i F D_i}{RT} \frac{\partial \phi}{\partial x} \frac{\partial c_i}{\partial x} + \frac{z_i D_i F c_i}{RT} \frac{\partial^2 \phi}{\partial x^2}$$

↓
i=1,2,3

$$\sum z_i c_i = 0$$

- Now, for making calculations simpler, we combine equations and add a $\sum z_i$.

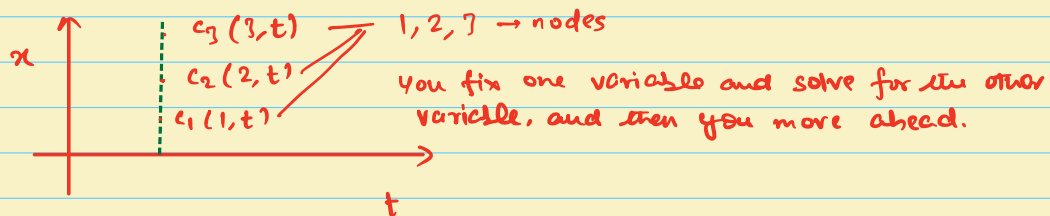
$$\sum z_i \left[\frac{\partial c_i}{\partial t} = D_i \frac{\partial^2 c_i}{\partial x^2} + \frac{z_i F D_i}{RT} \left(\frac{\partial c_i}{\partial x} \frac{\partial \phi}{\partial x} + c_i \frac{\partial^2 \phi}{\partial x^2} \right) \right]$$

$$\frac{\partial}{\partial t} (z_1 c_1 + z_2 c_2 + z_3 c_3) \xrightarrow{0}$$

$$0 = \sum z_i D_i \frac{\partial^2 c_i}{\partial x^2} + \sum \left[\frac{z_i^2 F D_i}{RT} \frac{\partial c_i}{\partial x} \right] \frac{\partial \phi}{\partial x} + \sum [z_i^2 c_i] \frac{\partial^2 \phi}{\partial x^2}$$

$$0 = \left[\sum z_i D_i \frac{\partial^2 c_i}{\partial x^2} \right] + \left[\sum z_i^2 \frac{F D_i}{RT} \frac{\partial c_i}{\partial x} \right] \frac{\partial \phi}{\partial x} + \left[\sum \frac{z_i^2 F D_i c_i}{RT} \right] \frac{\partial^2 \phi}{\partial x^2}$$

Now, we use - Method of Line



For an i^{th} space node,

$$\frac{\partial c_i}{\partial t} = D_i \left(\frac{c_{k+1} + c_{k-1} - 2c_k}{\Delta x^2} \right) + \frac{z_i F D_i}{RT} \left(\frac{c_{k+1} - c_{k-1}}{2\Delta x} \right) \left(\frac{\phi_{k+1} - \phi_{k-1}}{2\Delta x} \right)$$

$$+ \frac{z_i F D_i}{RT} c_k \left(\frac{\phi_{k+1} - \phi_{k-1} - 2\phi_k}{\Delta x^2} \right)$$

$$c_k = c_{i,k}$$

i: species
k: nodes for iteration

Now discretizing the last eq'n

$$0 = \sum z_i d_i \left(\frac{c_{i,k+1} + c_{i,k-1} - 2c_{i,k}}{\Delta x^2} \right) + \left[\sum z_i^2 \frac{FD_i}{RT} \left(\frac{c_{i,k+1} - c_{i,k-1}}{2\Delta x} \right) \right] \left(\frac{\phi_{i,k+1} - \phi_{i,k-1}}{2\Delta x} \right) + \left[\sum z_i^2 \frac{PD_i}{RT} c_{i,k} \right] \left(\frac{\phi_{i,k+1} + \phi_{i,k-1} - 2\phi_{i,k}}{\Delta x^2} \right)$$

Now, we use a DAE solver to solve these equations

At $t=0$, let all concentrations be 0.

✓
Zero-time sol'n.

Through zero-time sol'n, we get initial conditions.
Then we solve these on code.

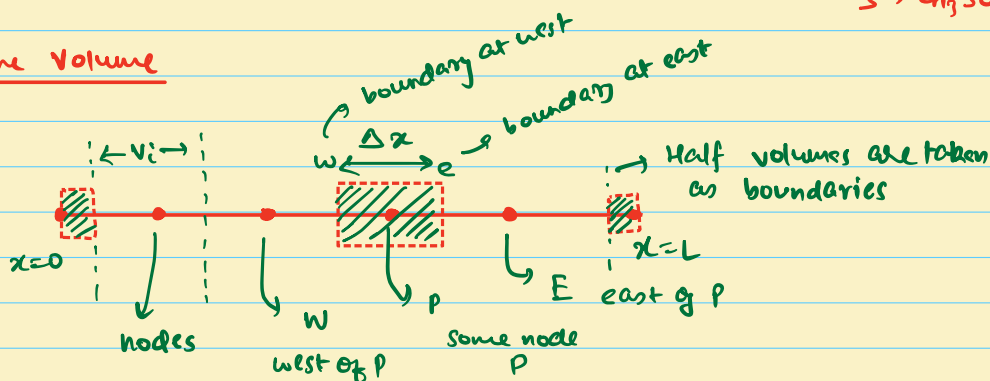
FINITE VOLUME TECHNIQUE FOR DISCRETIZATION OF PARTIAL DIFFERENTIAL EQUATION

$$\frac{\partial C_i}{\partial t} = D_i \frac{\partial^2 C_i}{\partial x^2} + \frac{z_i F D_i}{RT} \left[\frac{\partial C_i}{\partial x} \frac{\partial \phi}{\partial x} + C_i \frac{\partial^2 \phi}{\partial x^2} \right]$$

$$= D_i \frac{\partial^2 C_i}{\partial x^2} + \frac{\partial}{\partial x} \left[\frac{z_i F C_i D_i}{RT} \frac{\partial \phi}{\partial x} \right]$$

$$\begin{array}{l} 1 \rightarrow \text{Pb}^{+2} \\ 2 \rightarrow \text{H}^+ \\ 3 \rightarrow \text{CH}_3\text{SO}_3^- \end{array}$$

Finite Volume



$$\int_{\omega}^e \frac{\partial c_i}{\partial t} dx = \int_{\omega}^e D_i \frac{\partial^2 c_i}{\partial x^2} dx + \int_{\omega}^e \frac{\partial}{\partial x} \left[\frac{z_i c_i D_i}{RT} \frac{\partial \phi}{\partial x} \right] dx$$

Assuming c_i remain constant from $\omega \rightarrow e = c_{i,p}$

$$\frac{\partial}{\partial t} \int_{\omega}^e c_i dx = \left[\frac{d}{dt} c_{i,p} \right] \Delta x, \quad D_i \int_{\omega}^e \frac{\partial^2 c_i}{\partial x^2} dx = D_i \left[\frac{\partial c}{\partial x} \Big|_e - \frac{\partial c}{\partial x} \Big|_{\omega} \right]$$

$$\frac{z_i D_i c_i F}{RT} \int_{\omega}^e \frac{\partial \phi}{\partial x} dx = \frac{z_i D_i c_i F}{RT} \frac{\partial \phi}{\partial x} \Big|_e - \frac{z_i D_i c_i F}{RT} \frac{\partial \phi}{\partial x} \Big|_{\omega}$$

Hence,

$$\left[\frac{d}{dt} c_{i,p} \right] \Delta x = \text{Assuming all nodes are equally spaced.}$$

$$\frac{\partial c_i}{\partial x} \Big|_e = \frac{c_{i,e} - c_{i,p}}{\Delta x}$$

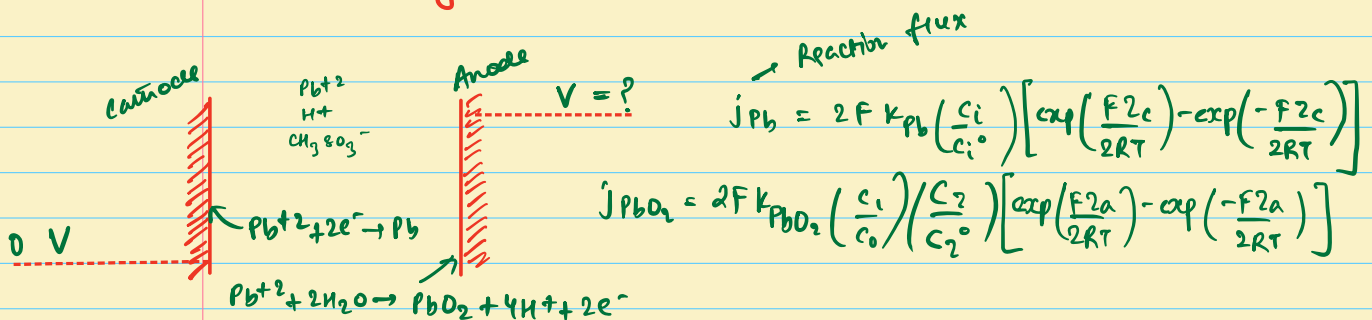
$$\frac{\partial c_i}{\partial x} \Big|_{\omega} = \frac{c_{i,p} - c_{i,\omega}}{\Delta x}$$

$$\frac{\partial \phi}{\partial x} \Big|_e = \frac{\phi_{i,e} - \phi_{i,p}}{\Delta x}$$

$$\frac{\partial \phi}{\partial x} \Big|_{\omega} = \frac{\phi_{i,p} - \phi_{i,\omega}}{\Delta x}$$

Let, $c_{i,e} = \frac{c_{i,p} + c_{i,e}}{2}$, then for consistency, $c_{i,\omega} = \frac{c_{i,\omega} + c_{i,p}}{2}$

For Boundary nodes, we Review BC.



$$Z_c = V_c - \phi - E_c$$

$$Z_a = V_a - \phi - E_a$$

Now, we can write

$$-D_1 \left. \frac{\partial C_1}{\partial x} \right|_p - \frac{z_1 C_1 D_1 F}{RT} \left. \frac{\partial \phi}{\partial x} \right|_p = -\frac{j_{Pb}}{2F}$$

$\swarrow$ Pb is the boundary $\swarrow$ Pb $\rightarrow$ Pb²⁺

@ $x=0$

Hence for B.C.

$$\left[\frac{d}{dt} C_{1,p} \right] \frac{\Delta x}{2} = D_1 \left. \frac{\partial C_1}{\partial x} \right|_e + \frac{z_1 C_1 D_1 F}{RT} \left. \frac{\partial \phi}{\partial x} \right|_e - D_1 \left. \frac{\partial C_1}{\partial x} \right|_p$$

$$\frac{-j_{Pb}}{2F} \Leftarrow -\frac{z_1 C_1 D_1 F}{RT}$$